

Master's Comprehensive Exam

Part I

Theory (90 minutes)

December 9, 2000

Instructions: Answer three out of four questions. Open book and open notes.

1. Let the joint mass function of the random vector (U, V) be as follows:

		v		
		0	1	2
u	3	.12	.05	.03
	4	.10	.25	.02
	5	.08	.05	.30

- Find the covariance of (U, V)
- Evaluate the c.d.f. $F_{U,V}(u, v)$ at $(u = 3.5, v = 4)$.
- Find the conditional expected value of V given $U=3$, i.e. $E[V | U = 3]$.

2. Let X_1, X_2, X_3 be binomial random variables. Let $X_1 \sim B(10, \frac{1}{2})$, $X_2 \sim B(10, \frac{1}{3})$, $X_3 \sim B(10, \frac{1}{4})$.

- Find the variance of $X_1 + 2X_2 - 8X_3 + 2$
- Find the second moment of X_3 .

3. Let X be a Poisson variable with parameter λ ; i.e. $f_X(x; \lambda) = e^{-\lambda} \lambda^x / x!$

- Test $H_0: \lambda = 1$ vs. $H_1: \lambda = 2$ at level of significance .0037. i.e. indicate for what values of X you reject and what values of X you do not reject.
- Find the power of the test.

4. Let X have density

$$f_X(x; \lambda) = \frac{1}{\lambda} e^{-x/\lambda} \quad x > 0.$$

Now consider a random sample of size n from this population. That is $X_1, \dots, X_n$.

- Find an unbiased estimator of λ .
- Find an unbiased estimator of $\lambda/3$.
- Are the above estimators consistent?
- Is the following estimator unbiased:
 $U = 0$ if $0 \leq X \leq 10$; $U = X$ if $X > 10$?